PEFT-Conformal FoV Prediction for 360° Video Streaming

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Introduction

One of the main issues in 360° video streaming arises from a mismatch between what is loaded into the buffer and what the user actually sees. A 360° video frame covers the entire sphere, but the player only displays a small portion determined by the user's head direction. Delivering every tile at a high resolution is expensive because it multiplies the bitrate by the number of tiles and places unnecessary load on both the network and the decoding pipeline. Yet failing to fetch the accurate region immediately degrades the user's Quality of Experience (QoE) as their view has not been fully loaded into the buffer, and the video quality is therefore lowered. This creates a challenging prediction problem: the system must anticipate the user's future Field of View (FoV) under tight real-time timing and bandwidth constraints and allocate resources accordingly.

This problem sits at the intersection of prediction, system design, and real-time decision-making. A VR streaming pipeline must interpret incoming head-motion signals, predict where the user will look next, and do it quickly enough to influence which tiles are fetched before playback deadlines. Solving this challenge is not only important for improving VR quality but also deeply relevant from a software engineering standpoint: building a system that can reliably operate under uncertainty, fluctuating network conditions, and tight latency constraints is exactly the kind of practice problems engineers tackle.

A major limitation of traditional FoV prediction is that user head movement is inherently uncertain, making any single point estimate fragile and easily invalidated by rapid or unexpected motion. Even a small prediction error can cause the system to fetch the wrong tiles, increasing bandwidth waste and noticeably reducing the viewer's QoE. This is why set-valued prediction matters: instead of committing to a single future viewing point, the model predicts a calibrated region of plausible FoV positions that is statistically guaranteed to contain the truth with the probability of at least $1 - \alpha$. By working with sets rather than points, the streaming pipeline has increased capability to handle uncertainty, enabling smoother playback and more reliable tile allocation even during highly dynamic user behavior.

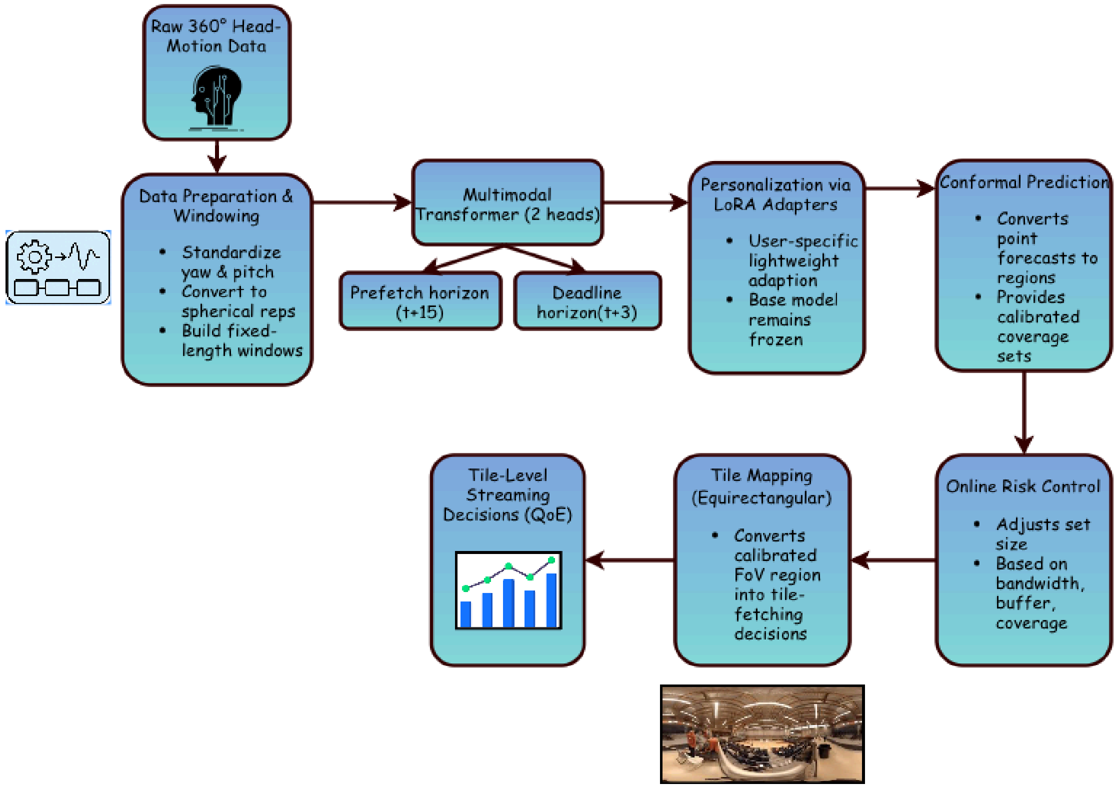


Figure 1. Complete pipeline architecture and data flow between components

System Architecture

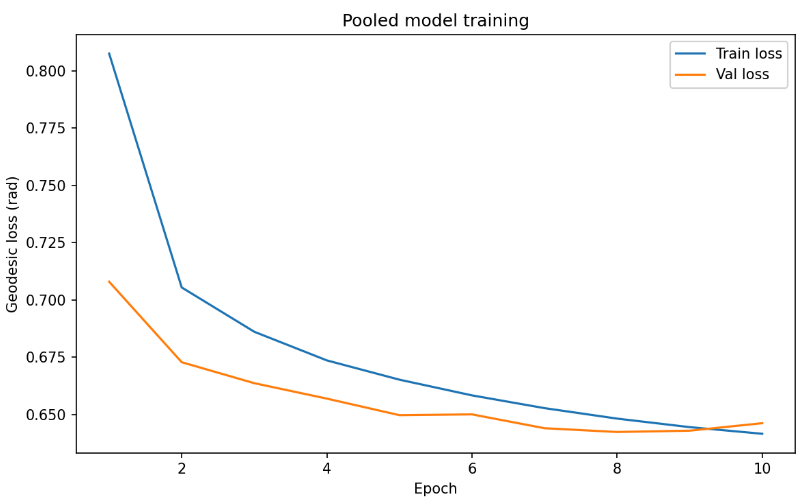
Our system is structured as an end-to-end pipeline that begins with raw 360° head-motion and ends with tile-level streaming decisions driven by calibrated FoV predictions. The first stage prepared the motion data by standardizing yaw and pitch signals, converting them into spherical representations, and forming fixed-length temporal windows that capture recent head-movement history.

At the core of the pipeline is a multi-horizon Transformer with two output heads corresponding to the prefetch and near-deadline horizons. The model interprets a window of past orientations and produces forecasts of where the user is likely to look at both future times. To account for differences in individual behavior, the Transformer is further adapted through parameter-efficient personalization, with LoRA adapters fine-tuned for each user. LoRA adapters are used in particular to make this personalization cost-efficient in storage and computation. Without these adapters, we could not meet many of the real world constraints as many devices that play 360° videos cannot quickly recompute the approximately 7 million parameters of the

Transformer required to retrain the model to personalize the predictions. These calculations would also require a lot of energy, and energy consumption needs to be minimized as many of these devices also run on limited battery life. This two-stage training approach enables the system to learn general motion patterns while also adjusting to user-specific tendencies without retraining the full model.

To handle prediction uncertainty, the system then applies split conformal prediction to convert each point forecast to a calibrated region on the sphere. These regions are designed so that the true future FoV center falls within them at a controlled frequency determined by the target coverage level (1 - α). During runtime, an online risk controller adjusts how large or small these regions should be by responding to current bandwidth, buffer levels, and recent coverage performance.

The final stage of the pipeline maps each calibrated region to a set of video tiles in the equirectangular projection, allowing the streaming system to decide which tiles should be fetched at higher quality.



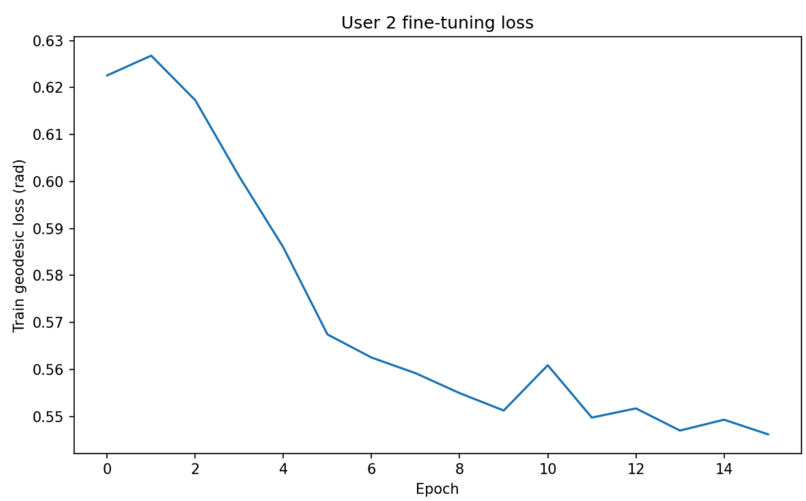


Figure 2. (Left)Training and validation geodesic loss (in radians) for the pooled base model over 10 epochs. Loss is the sum of prefetch and deadline prediction errors.

Figure 3. (Right)Geodesic loss (in radians) for fine-tuning the LoRA adapter on User 2. The loss is computed as the sum of prefetch and deadline loss values.

Dataset And Evaluation

We use the AVTrack360 dataset, which contains head-tracking trajectories from users watching 360° videos. The dataset provides yaw and pitch angles sampled at ~10Hz (0.1s intervals) across multiple videos and users. The data was also cleaned through padding and linear interpolation to ensure uniform 0.1-second timesteps across all videos and users. We convert angles to radians and use spherical geometry (geodesic distance) throughout the pipeline. Temporal sequences are segmented into 15-timestep context windows, with care taken to prevent windows from spanning video boundaries.

To ensure meaningful evaluation, the dataset was structured to separate training, calibration, and testing across both users and time. The pooled AVTrack360 dataset was randomly divided into an 80/20 user-split, with the base Transformer trained only on users in the training partition. Personalization and end-to-end testing were then performed exclusively on users from the validation set, ensuring that the core model did not encounter these users during pooled training. For each validation user, their sequence was further divided chronologically into distinct segments for personalization via LoRA adapter training, conformal calibration, and evaluation. This multi-stage split provides a clean separation between learning, calibration, and testing, enabling a reliable assessment of how well personalization and calibrated prediction improve performance on previously unseen users.

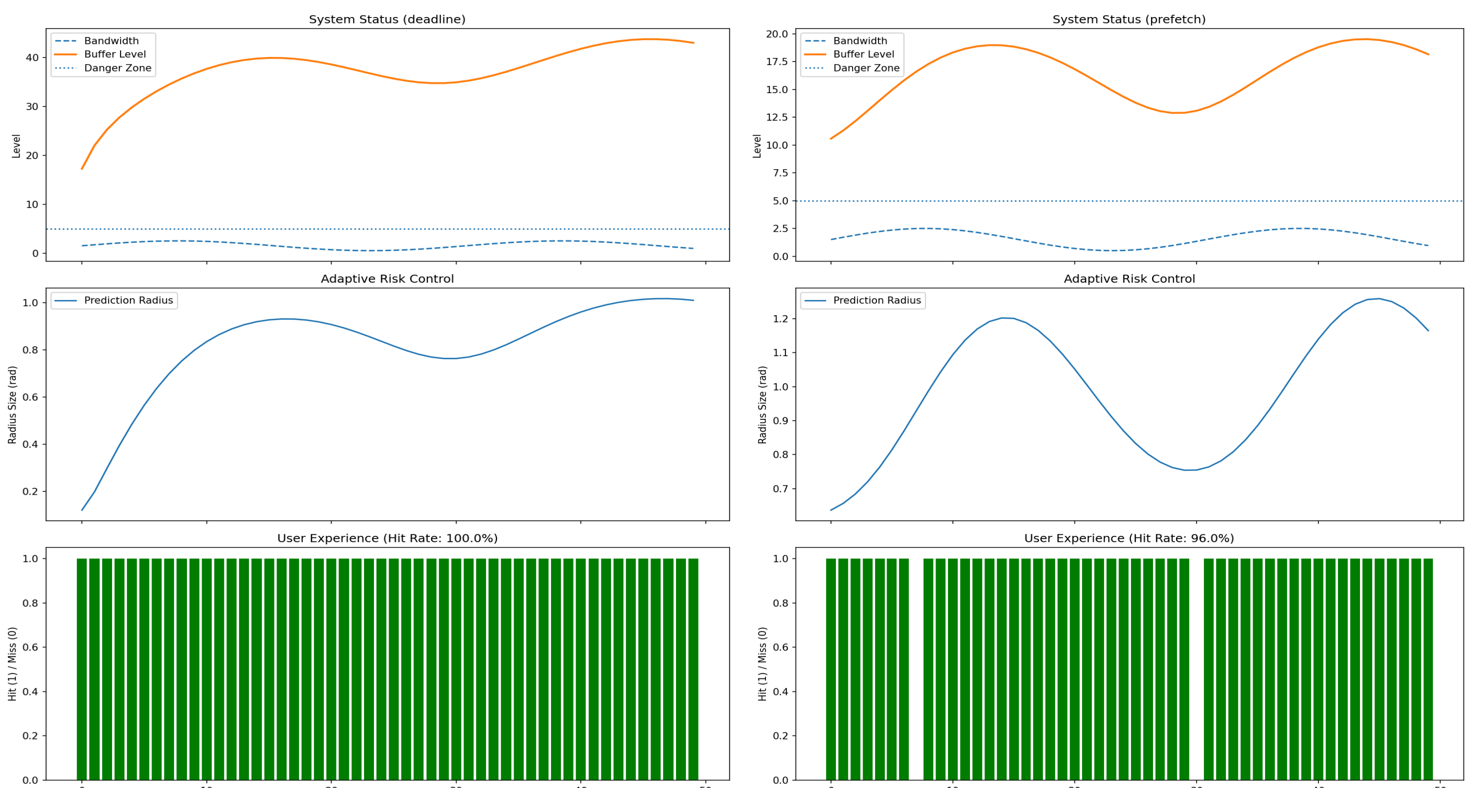


Figure 4. System performance under variable bandwidth for prefetch (left) and deadline (right) horizons for user 2. Top: bandwidth and buffer levels; Middle: adaptive radius adjustment; Bottom: tile hit/miss events over 50 timesteps.

Coverage was assessed by measuring how often the ground-truth viewport tile fell within the prefetched tile set. We also tracked buffer stability (maintaining levels above 5s danger zone), computational overhead (inference latency per frame), and adaptive behavior (α and radius adjustment in response to bandwidth fluctuations). Simulations ran for 50 timesteps under synthetic variable bandwidth conditions (sinusoidal pattern ranging 0.5-2.5 units).

Results

The system achieved approximately 96-98% hit rate at the prefetch horizon and 100% at the near-deadline horizon, indicating that the conformal prediction radii were appropriately calibrated for both long and short-range forecasting. The adaptive α control responded effectively to bandwidth fluctuations by contracting the radius during periods of low buffer availability and expanding it when network conditions improved, allowing the system to maintain accurate coverage without compromising playback stability.

The computational overhead of the pipeline was also evaluated. Conformal radii are computed once from the calibration segment and do not contribute to runtime latency. During each frame of the simulation, the system performs only a small number of lightweight operations: Transformer inference on a short context window, radius adjustment based on the controller signal, and geometric tile selection. These steps are computationally inexpensive and remain well within real-time constraints. The α-controller imposes negligible additional cost, as it updates only a single scalar parameter per frame, and LoRA-based personalization does not affect inference speed once the adapters are loaded. Overall, the latency and overhead introduced by the model, calibration, and control components are minimal, supporting the feasibility of this design for real-time VR streaming applications.

Discussion

Throughout the development and evaluation of the PEFT-Personalized Conformal 360° Fov predictor, several key lessons emerged, highlighting both the strengths and limitations of our approach. First, we confirmed that personalization via PEFT–specifically LoRA adapters–can significantly enhance prediction accuracy for individual users. However, this benefit is contingent on the availability of per-user data, introducing a cold-start problem where new users must rely on the pooled model until sufficient personalization data is collected. This underscores a fundamental tension between adaptability and immediacy in personalized systems. Second, conformal predictions proved to be a robust framework for providing uncertainty-aware set-valued forecasts. Yet, we observed that set size must be carefully managed: while larger areas improve coverage, they also increase bandwidth consumption. This trade-off necessitates dynamic control, which our α controller handled well under moderate network variations but struggled with during extreme bandwidth drops or sudden, unpredictable viewport shifts. Finally, system integration required careful attention to latency budgets, ensuring that prediction, calibration, and tile selection all complete within frame deadlines (~100ms for 10Hz playback). These benefits and limitations collectively emphasize that while the combination of PEFT and CP offers a principled pathway to risk-aware personalization, real-world deployment requires robust fallback mechanisms, rigorous latency profiling, and continuous adaptation to changing user and network conditions.

Future Work

Looking ahead, several directions for future work could extend the capabilities and practicality of our system. One of the opportunities is to enrich the prediction model with visual saliency features, which would allow content-aware forecasting, especially in dynamic scenes where head motion alone is insufficient. On the control side, more sophisticated strategies such as Reinforcement Learning could be explored to optimize α and tile quality selection, so that our system can anticipate network fluctuations and user interactions better. From a development perspective, future work should address scalability and fairness in multi-user environments, such as developing energy-efficient inference for mobile devices and incorporating subjective quality-of-experience assessments through user studies. Finally, testing in real-world network conditions–including mobile, Wi-Fi, and broadband–will be essential to validate the system's performance and inform future refinements. Together, these advancements would push the system toward a practical and deployable solution that balances personalization, risk control, and resource efficiency in real time.

Conclusion

We developed a personalized, uncertainty-aware FoV prediction system for 360° video streaming that integrates parameter-efficient fine-tuning, split conformal prediction, and adaptive risk control. Our results show that combining lightweight personalization with principled uncertainty quantification leads to highly reliable tile selection, achieving 96–98% hit rates for prefetch decisions and 100% coverage near the playback deadline. Importantly, this performance was maintained without violating real-time latency constraints, demonstrating that the approach is computationally feasible for practical VR streaming systems.

More broadly, this model illustrates that accurate FoV prediction is not only a modeling problem but a systems problem involving uncertainty, bandwidth constraints, and strict timing requirements. By addressing all three, the proposed design provides a pathway for scalable and robust 360° video delivery. PEFT, conformal calibration, and online control together can form a general framework that can extend to future VR/AR streaming pipelines where personalization and reliability coexist under real-world conditions.